Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: PH RANCH #3

Physical address of dairy:

Number and Street	City	County	Zip Code
11586 N Shaffer RD	Winton	Merced	95388

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 09/27/1973

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0042-0260-0023-0000

B. OPERATORS

Bettencourt, Mike

Operator name: Bettencourt, Mike

Telephone no.: (209) 358-2328 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11570 N Shaffer AVE	Winton	CA	95388

Van Warmerdam, Paul

Operator name: Van Warmerdam, Paul

Telephone no.: (209) 358-5111 (Landline) (209) 678-2251 (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6335 W Oakdale RD	Winton	CA	95388

C. OWNERS

Bettencourt, Mike

Legal owner name: Bettencourt, Mike

Telephone no.: (209) 358-2328 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
11570 N Shaffer AVE	Winton	CA	95388

Bettencourt Family Property

Legal owner name:	Bettencourt Family Property	Telephone no.:	(209) 358-5129	
			Landline	Cellular
11570 N Shaffer Avenue	Winton		CA	95388
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

McDowell, Lorrie

Legal owner name:	McDowell, Lorrie	Telephone no.:	(209) 394-1601	(209) 394-1601
			Landline	Cellular
11576 Shaffer Road	Winton		CA	95388
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	450	0	25	0	0	0
Maximum number	480	0	75	0	0	0
Average number	450	0	25	0	0	0
Avg live weight (lbs)	1,400	0	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,698.24 *tons per reporting period*

Total nitrogen from manure: 151,912.53 *lbs per reporting period*

After ammonia losses (30% loss applied): 106,338.77 *lbs per reporting period*

Total phosphorus from manure: 25,697.59 *lbs per reporting period*

Total potassium from manure: 85,587.85 *lbs per reporting period*

Total salt from manure: 211,882.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 24,192,000 *gallons*

Total nitrogen generated: 50,075.61 *lbs*

Total phosphorus generated: 11,083.36 *lbs*

Total potassium generated: 97,461.86 *lbs*

Total salt generated: 592,171.88 *lbs*

	24,192,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	24,192,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Barn Dom.	Ground water
D-1	Ground water
D-4	Ground water
D-5	Ground water
D-6	Ground water

Source Description	Type
D-7	Ground water
DW-1	Ground water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Fd #1,2	76	76	3	both	0042-0140-0028-0000
Fd #10	22	22	0	none	0042-0260-0015-0000
Fd #11	9	9	1	process wastewater	0042-0260-0026-0000
Fd #14	25	25	0	none	0042-0250-0005-0000
Fd #1a/2a	14	14	1	process wastewater	0042-0260-0019-0000
Fd #3	29	29	2	both	0042-0140-0028-0000
Fd #3a	37	37	0	none	0042-0260-0021-0000
Fd #5	29	29	0	none	0042-0250-0005-0000
Fd #6	94	94	3	both	0042-0140-0028-0000
Fd #7	70	70	2	both	0042-0260-0025-0000
Fd #8,9	40	40	0	none	0042-0260-0025-0000
Totals for areas that were used for application	292	292	12		
Totals for areas that were not used for application	153	153	0		
Land application area totals	445	445	12		

B. CROPS AND HARVESTS

Fd #1,2

Field name: Fd #1,2

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 76 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	652.00 *ton*	Dry-weight		69.8	10,800.00	2,250.00	15,400.00		9.60
05/25/2023	36.00 *ton*	Dry-weight		6.2	22,500.00	2,700.00	11,500.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	832.00
Total actual harvest content	9.05	75.96	14.06	90.02	592.53

Fd #1,2

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,955.00 *ton*	Dry-weight		72.8	11,300.00	1,820.00	9,630.00		3.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	25.72	158.13	25.47	134.76	433.80

Fd #11

Field name: Fd #11

01/02/2016: Pasture Grass

Crop: Pasture Grass Acres planted: 9 Plant date: 01/02/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/15/2023	126.00 *ton*	Dry-weight		65.0	31,500.00	2,680.00	15,400.00		19.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	720.00	22.80	504.00	1,020.00
Total actual harvest content	14.00	308.70	26.26	150.92	1,911.00

Fd #1a/2a

Field name: Fd #1a/2a

Fd #1a/2a

01/02/2016: Pasture Grass

Crop: Pasture Grass Acres planted: 14 Plant date: 01/02/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/15/2023	196.00 *ton*	Dry-weight		65.0	31,500.00	2,680.00	15,400.00		19.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	720.00	22.80	504.00	1,020.00
Total actual harvest content	14.00	308.70	26.26	150.92	1,911.00

Fd #3

Field name: Fd #3

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	591.00 *ton*	Dry-weight		74.6	17,400.00	2,930.00	20,100.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	832.00
Total actual harvest content	20.38	180.14	30.33	208.09	1,221.62

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	760.00 *ton*	Dry-weight		70.4	11,300.00	2,010.00	10,400.00		3.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	26.21	175.31	31.18	161.35	496.46

Fd #6

Field name: Fd #6

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 94 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	1,916.00 *ton*	Dry-weight		78.0	20,000.00	3,020.00	21,700.00		9.70
05/25/2023	45.00 *ton*	Dry-weight		6.2	22,500.00	2,700.00	11,500.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	832.00
Total actual harvest content	20.86	199.58	29.51	204.94	966.04

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 94 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	2,343.00 *ton*	Dry-weight		68.8	10,100.00	2,340.00	10,400.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	24.93	157.09	36.40	161.76	855.44

Fd #7

Field name: Fd #7

Fd #7

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 70 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	2,078.00 *ton*	Dry-weight		69.2	12,300.00	2,170.00	15,100.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	832.00
Total actual harvest content	29.69	224.92	39.68	276.12	1,590.92

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,777.00 *ton*	Dry-weight		66.8	9,400.00	2,280.00	10,000.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	252.00	42.00	184.80	1,120.00
Total actual harvest content	25.39	158.45	38.43	168.56	927.09

NUTRIENT BUDGET

A. LAND APPLICATIONS

Fd #1,2 - 10/25/2022: Oats, silage-soft dough

Field name: Fd #1,2

Crop: Oats, silage-soft dough — Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	32.89	13.33	150.27	789.31	2,304,000.00 *gal*
River	Surface water	0.69	0.00	0.00	66.82	4,140,000.00 *gal*
Application event totals		33.57	13.33	150.27	856.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	39.63	7.62	63.50	443.57	768,000.00 *gal*
Application event totals		39.63	7.62	63.50	443.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	19.82	3.81	31.75	221.78	384,000.00 *gal*
Application event totals		19.82	3.81	31.75	221.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.48	0.00	0.00	144.40	8,946,000.00 *gal*
Application event totals		1.48	0.00	0.00	144.40	

Fd #1,2 - 06/30/2023: Corn, silage

Field name: Fd #1,2

Crop: Corn, silage — Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Fd #1,2 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Application	Corral solids	171.69	52.04	187.58	0.00	532.00 *ton*
Application event totals		171.69	52.04	187.58	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	18.89	4.59	31.29	121.43	768,000.00 *gal*
River	Surface water	1.43	0.00	0.00	139.46	8,640,000.00 *gal*
Application event totals		20.32	4.59	31.29	260.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.44	0.00	0.00	140.33	8,694,000.00 *gal*
Application event totals		1.44	0.00	0.00	140.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.48	0.00	0.00	144.40	8,946,000.00 *gal*
Application event totals		1.48	0.00	0.00	144.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	19.86	1.00	4.81	134.08	1,152,000.00 *gal*
River	Surface water	1.46	0.00	0.00	142.36	8,820,000.00 *gal*
Application event totals		21.32	1.00	4.81	276.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.41	0.00	0.00	137.42	8,514,000.00 *gal*
Application event totals		1.41	0.00	0.00	137.42	

Fd #1,2 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.46	0.00	0.00	142.36	8,820,000.00 *gal*
Application event totals		1.46	0.00	0.00	142.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.48	0.00	0.00	144.40	8,946,000.00 *gal*
Application event totals		1.48	0.00	0.00	144.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.43	0.00	0.00	139.46	8,640,000.00 *gal*
Application event totals		1.43	0.00	0.00	139.46	

Fd #11 - 01/02/2016: Pasture Grass

Field name: Fd #11

Crop: Pasture Grass

Plant date: 01/02/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.51	0.00	0.00	147.21	1,080,000.00 *gal*
Application event totals		1.51	0.00	0.00	147.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	39.88	9.68	66.05	256.36	192,000.00 *gal*
River	Surface water	1.01	0.00	0.00	98.14	720,000.00 *gal*
Application event totals		40.89	9.68	66.05	354.50	

Fd #11 - 01/02/2016: Pasture Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.59	0.00	0.00	154.57	1,134,000.00 *gal*
Application event totals		1.59	0.00	0.00	154.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	59.82	14.53	99.07	384.54	288,000.00 *gal*
River	Surface water	1.01	0.00	0.00	98.14	720,000.00 *gal*
Application event totals		60.83	14.53	99.07	482.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.76	0.00	0.00	171.74	1,260,000.00 *gal*
Application event totals		1.76	0.00	0.00	171.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	41.93	2.11	10.15	283.06	288,000.00 *gal*
River	Surface water	1.01	0.00	0.00	98.14	720,000.00 *gal*
Application event totals		42.93	2.11	10.15	381.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.51	0.00	0.00	147.21	1,080,000.00 *gal*
Application event totals		1.51	0.00	0.00	147.21	

Fd #1a/2a - 01/02/2016: Pasture Grass

Field name: Fd #1a/2a

Crop: Pasture Grass

Plant date: 01/02/2016

Fd #1a/2a - 01/02/2016: Pasture Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	76.91	18.68	127.38	494.41	576,000.00 *gal*
DW-1	Ground water	5.16	0.00	0.00	130.41	990,000.00 *gal*
Application event totals		82.07	18.68	127.38	624.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
DW-1	Ground water	8.50	0.00	0.00	214.72	1,630,000.00 *gal*
Application event totals		8.50	0.00	0.00	214.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	76.91	18.68	127.38	494.41	576,000.00 *gal*
DW-1	Ground water	5.16	0.00	0.00	130.41	990,000.00 *gal*
Application event totals		82.07	18.68	127.38	624.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
DW-1	Ground water	8.61	0.00	0.00	217.36	1,650,000.00 *gal*
Application event totals		8.61	0.00	0.00	217.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	71.87	3.61	17.40	485.25	768,000.00 *gal*
DW-1	Ground water	5.16	0.00	0.00	130.41	990,000.00 *gal*
Application event totals		77.04	3.61	17.40	615.66	

Fd #1a/2a - 01/02/2016: Pasture Grass

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
DW-1	Ground water	8.61	0.00	0.00	217.36	1,650,000.00 *gal*
Application event totals		8.61	0.00	0.00	217.36	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
DW-1	Ground water	8.61	0.00	0.00	217.36	1,650,000.00 *gal*
Application event totals		8.61	0.00	0.00	217.36	

Fd #3 - 10/26/2022: Oats, silage-soft dough

Field name: Fd #3

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	28.73	11.65	131.27	689.52	768,000.00 *gal*
River	Surface water	1.09	0.00	0.00	106.60	2,520,000.00 *gal*
Application event totals		29.82	11.65	131.27	796.11	

02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	51.93	9.99	83.21	581.23	384,000.00 *gal*
Application event totals		51.93	9.99	83.21	581.23	

03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	51.93	9.99	83.21	581.23	384,000.00 *gal*
Application event totals		51.93	9.99	83.21	581.23	

Fd #3 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.41	0.00	0.00	137.05	3,240,000.00 *gal*
Application event totals		1.41	0.00	0.00	137.05	

Fd #3 - 06/15/2023: Corn, silage

Field name: Fd #3

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Application	Corral solids	171.69	52.04	187.58	0.00	203.00 *ton*
Application event totals		171.69	52.04	187.58	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	49.50	12.02	81.99	318.24	768,000.00 *gal*
River	Surface water	1.25	0.00	0.00	121.83	2,880,000.00 *gal*
Application event totals		50.76	12.02	81.99	440.06	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.43	0.00	0.00	139.34	3,294,000.00 *gal*
Application event totals		1.43	0.00	0.00	139.34	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.38	0.00	0.00	134.77	3,186,000.00 *gal*
Application event totals		1.38	0.00	0.00	134.77	

Fd #3 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.49	0.00	0.00	144.67	3,420,000.00 *gal*
Application event totals		1.49	0.00	0.00	144.67	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.56	0.00	0.00	152.28	3,600,000.00 *gal*
Application event totals		1.56	0.00	0.00	152.28	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.59	0.00	0.00	154.57	3,654,000.00 *gal*
Application event totals		1.59	0.00	0.00	154.57	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.64	0.00	0.00	159.90	3,780,000.00 *gal*
Application event totals		1.64	0.00	0.00	159.90	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.64	0.00	0.00	159.90	3,780,000.00 *gal*
Application event totals		1.64	0.00	0.00	159.90	

10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.67	0.00	0.00	162.18	3,834,000.00 *gal*
Application event totals		1.67	0.00	0.00	162.18	

Fd #6 - 10/27/2022: Oats, silage-soft dough

Fd #6 - 10/27/2022: Oats, silage-soft dough

Field name: Fd #6

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	39.89	16.17	182.25	957.25	3,456,000.00 *gal*
River	Surface water	1.32	0.00	0.00	128.81	9,870,000.00 *gal*
Application event totals		41.21	16.17	182.25	1,086.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	48.07	9.25	77.01	537.94	1,152,000.00 *gal*
Application event totals		48.07	9.25	77.01	537.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	48.07	9.25	77.01	537.94	1,152,000.00 *gal*
Application event totals		48.07	9.25	77.01	537.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.44	0.00	0.00	139.77	10,710,000.00 *gal*
Application event totals		1.44	0.00	0.00	139.77	

Fd #6 - 06/15/2023: Corn, silage

Field name: Fd #6

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Fd #6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Application	Corral solids	159.42	48.32	174.19	0.00	611.00 *ton*
Application event totals		159.42	48.32	174.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.44	0.00	0.00	139.77	10,710,000.00 *gal*
Application event totals		1.44	0.00	0.00	139.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	22.91	5.56	37.94	147.27	1,152,000.00 *gal*
River	Surface water	1.27	0.00	0.00	123.32	9,450,000.00 *gal*
Application event totals		24.18	5.56	37.94	270.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.47	0.00	0.00	143.33	10,983,000.00 *gal*
Application event totals		1.47	0.00	0.00	143.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.44	0.00	0.00	139.77	10,710,000.00 *gal*
Application event totals		1.44	0.00	0.00	139.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.46	0.00	0.00	141.69	10,857,000.00 *gal*
Application event totals		1.46	0.00	0.00	141.69	

Fd #6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.46	0.00	0.00	142.51	10,920,000.00 *gal*
Application event totals		1.46	0.00	0.00	142.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	16.06	0.81	3.89	108.41	1,152,000.00 *gal*
River	Surface water	1.11	0.00	0.00	107.70	8,253,000.00 *gal*
Application event totals		17.16	0.81	3.89	216.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.42	0.00	0.00	137.85	10,563,000.00 *gal*
Application event totals		1.42	0.00	0.00	137.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.41	0.00	0.00	137.03	10,500,000.00 *gal*
Application event totals		1.41	0.00	0.00	137.03	

Fd #7 - 10/28/2022: Oats, silage-soft dough

Field name: Fd #7

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Fd #7 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	17.85	7.24	81.58	428.48	1,152,000.00 *gal*
River	Surface water	1.20	0.00	0.00	116.71	6,660,000.00 *gal*
Application event totals		19.05	7.24	81.58	545.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	96.82	18.62	155.12	1,083.57	1,728,000.00 *gal*
Application event totals		96.82	18.62	155.12	1,083.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	32.27	6.21	51.71	361.19	576,000.00 *gal*
Application event totals		32.27	6.21	51.71	361.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.59	0.00	0.00	154.57	8,820,000.00 *gal*
Application event totals		1.59	0.00	0.00	154.57	

Fd #7 - 06/11/2023: Corn, silage

Field name: Fd #7

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.75	0.00	0.00	170.34	9,720,000.00 *gal*
Application event totals		1.75	0.00	0.00	170.34	

Fd #7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost Application	Corral solids	145.76	44.18	159.26	0.00	416.00 *ton*
Application event totals		145.76	44.18	159.26	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.68	0.00	0.00	164.03	9,360,000.00 *gal*
Application event totals		1.68	0.00	0.00	164.03	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Application	Process wastewater	21.56	1.08	5.22	145.57	1,152,000.00 *gal*
River	Surface water	1.31	0.00	0.00	127.12	7,254,000.00 *gal*
Application event totals		22.87	1.08	5.22	272.70	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.68	0.00	0.00	164.03	9,360,000.00 *gal*
Application event totals		1.68	0.00	0.00	164.03	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Blended Lg and Fresh Water Application	Process wastewater	30.76	7.47	50.95	197.76	1,152,000.00 *gal*
River	Surface water	1.23	0.00	0.00	119.87	6,840,000.00 *gal*
Application event totals		31.99	7.47	50.95	317.63	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.68	0.00	0.00	164.03	9,360,000.00 *gal*
Application event totals		1.68	0.00	0.00	164.03	

Fd #7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.58	0.00	0.00	153.62	8,766,000.00 *gal*
Application event totals		1.58	0.00	0.00	153.62	

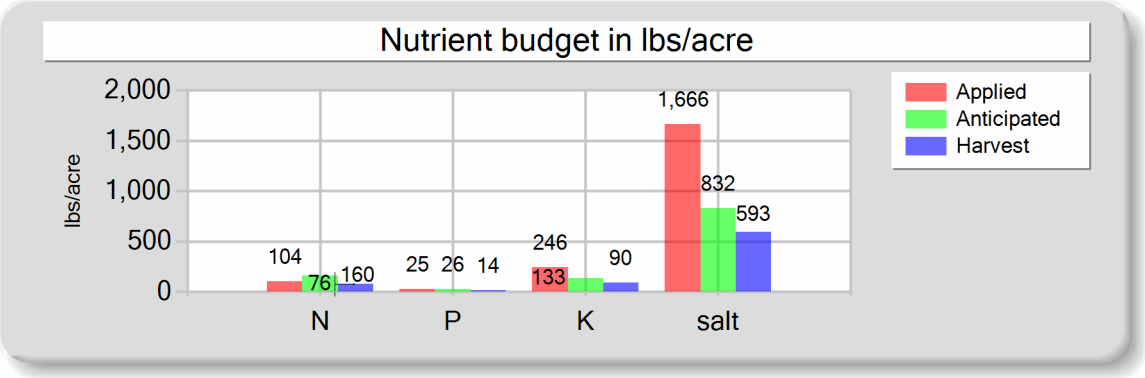
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	1.56	0.00	0.00	151.41	8,640,000.00 *gal*
Application event totals		1.56	0.00	0.00	151.41	

B. NUTRIENT BUDGET

Fd #1,2 - 10/25/2022: Oats, silage-soft dough

Field name: Fd #1,2 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	92.34	24.77	245.52	1,454.67
Fresh water	2.17	0.00	0.00	211.22
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	103.84	24.77	245.52	1,665.89
Anticipated crop nutrient removal	160.00	25.60	132.80	832.00
Actual crop nutrient removal	75.96	14.06	90.02	592.53
Nutrient balance	27.89	10.71	155.50	1,073.36
Applied to removed ratio	1.37	1.76	2.73	2.81

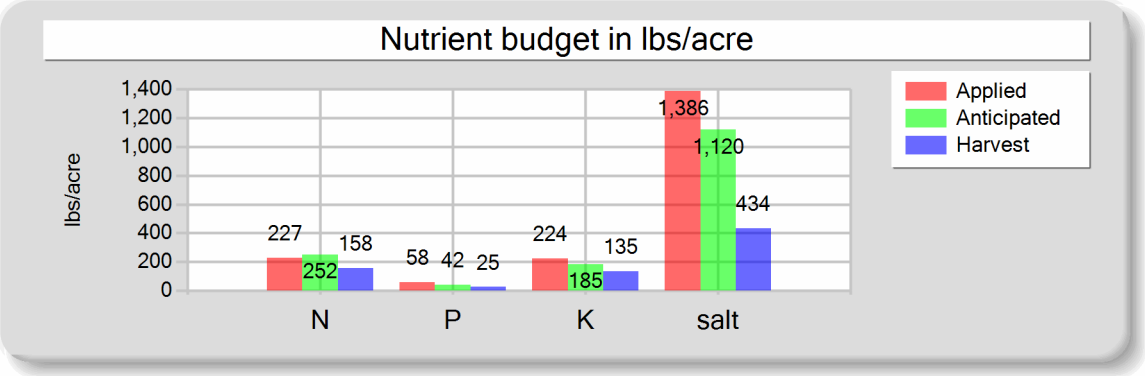
Fresh water applied	
13,086,000.00	*gallons*
481.91	*acre-inches*
6.34	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
1.67	*inches/acre*

Total harvests for the crop	
2	*harvests*

Fd #1,2 - 06/30/2023: Corn, silage

Field name: Fd #1,2 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	171.69	52.04	187.58	0.00
Process wastewater	38.75	5.59	36.09	255.52
Fresh water	11.61	0.00	0.00	1,130.19
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	226.71	57.62	223.68	1,385.71
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	158.13	25.47	134.76	433.80
Nutrient balance	68.58	32.15	88.92	951.90
Applied to removed ratio	1.43	2.26	1.66	3.19

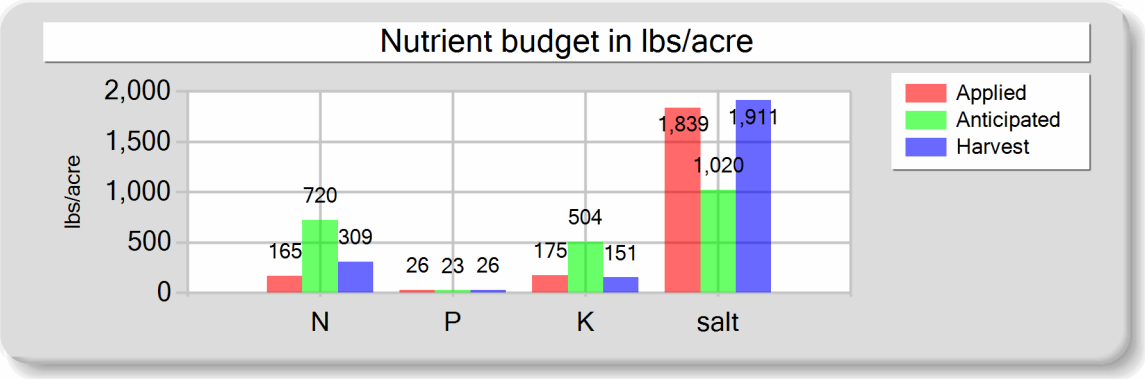
Fresh water applied	
70,020,000.00	*gallons*
2,578.60	*acre-inches*
33.93	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #11 - 01/02/2016: Pasture Grass

Field name: Fd #11 Crop: Pasture Grass Plant date: 01/02/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	141.62	26.32	175.27	923.96
Fresh water	9.40	0.00	0.00	915.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	165.02	26.32	175.27	1,839.09
Anticipated crop nutrient removal	720.00	22.80	504.00	1,020.00
Actual crop nutrient removal	308.70	26.26	150.92	1,911.00
Nutrient balance	-143.68	0.05	24.35	-71.91
Applied to removed ratio	0.53	1.00	1.16	0.96

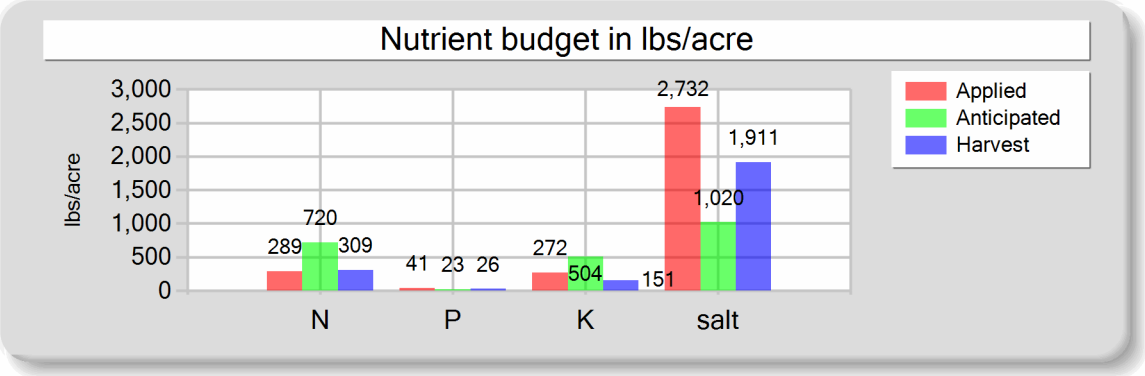
Fresh water applied	
6,714,000.00	*gallons*
247.25	*acre-inches*
27.47	*inches/acre*

Process wastewater applied	
768,000.00	*gallons*
28.28	*acre-inches*
3.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #1a/2a - 01/02/2016: Pasture Grass

Field name: Fd #1a/2a Crop: Pasture Grass Plant date: 01/02/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.69	40.97	272.15	1,474.06
Fresh water	49.81	0.00	0.00	1,258.04
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	289.50	40.97	272.15	2,732.10
Anticipated crop nutrient removal	720.00	22.80	504.00	1,020.00
Actual crop nutrient removal	308.70	26.26	150.92	1,911.00
Nutrient balance	-19.20	14.70	121.23	821.10
Applied to removed ratio	0.94	1.56	1.80	1.43

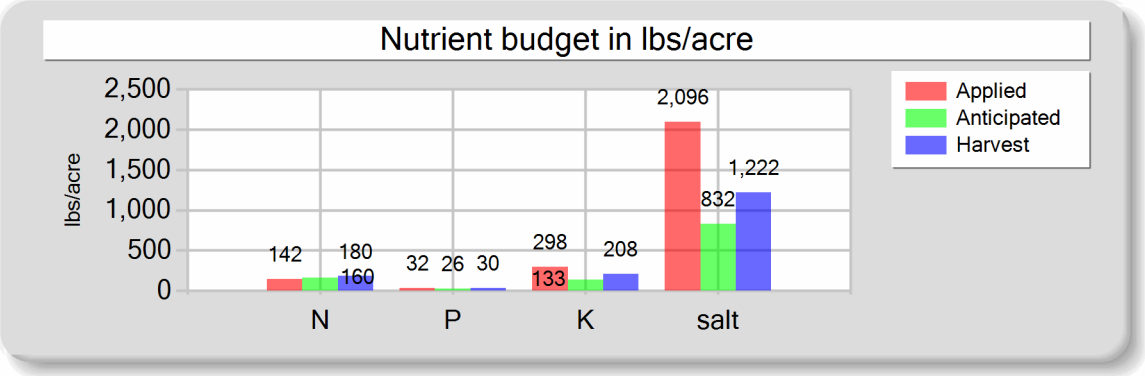
Fresh water applied	
9,550,000.00	*gallons*
351.69	*acre-inches*
25.12	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
5.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #3 - 10/26/2022: Oats, silage-soft dough

Field name: Fd #3 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	132.60	31.62	297.69	1,851.97
Fresh water	2.50	0.00	0.00	243.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	142.10	31.62	297.69	2,095.62
Anticipated crop nutrient removal	160.00	25.60	132.80	832.00
Actual crop nutrient removal	180.14	30.33	208.09	1,221.62
Nutrient balance	-38.03	1.29	89.60	874.00
Applied to removed ratio	0.79	1.04	1.43	1.72

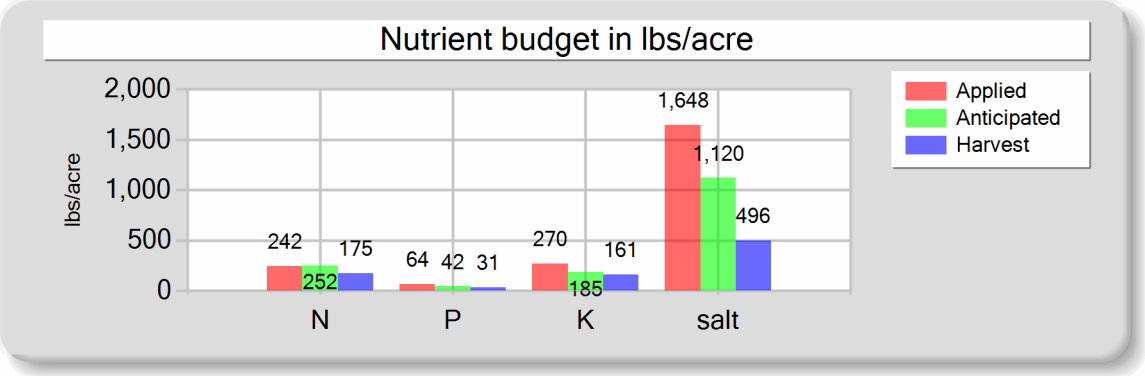
Fresh water applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
7.31	*inches/acre*

Process wastewater applied	
1,536,000.00	*gallons*
56.57	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #3 - 06/15/2023: Corn, silage

Field name: Fd #3 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	171.69	52.04	187.58	0.00
Process wastewater	49.50	12.02	81.99	318.24
Fresh water	13.66	0.00	0.00	1,329.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.85	64.06	269.58	1,647.66
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	175.31	31.18	161.35	496.46
Nutrient balance	66.53	32.87	108.22	1,151.20
Applied to removed ratio	1.38	2.05	1.67	3.32

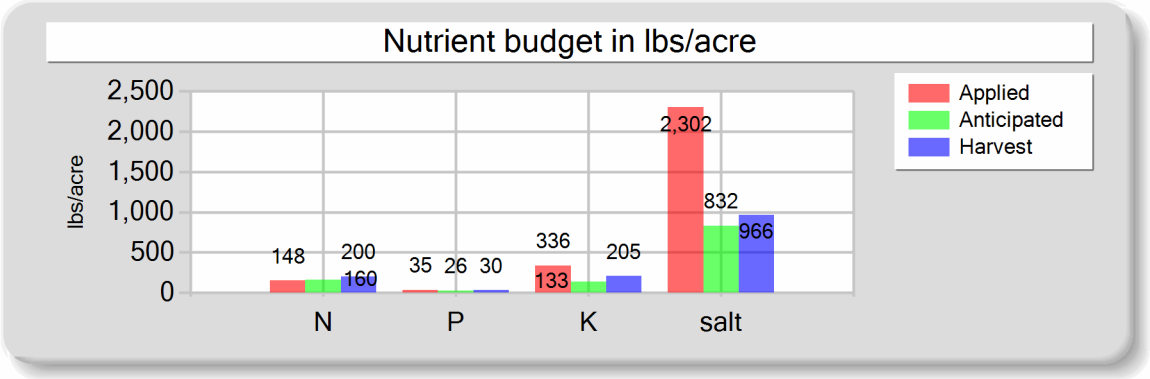
Fresh water applied	
31,428,000.00	*gallons*
1,157.39	*acre-inches*
39.91	*inches/acre*

Process wastewater applied	
768,000.00	*gallons*
28.28	*acre-inches*
0.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #6 - 10/27/2022: Oats, silage-soft dough

Field name: Fd #6 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	136.02	34.66	336.27	2,033.14
Fresh water	2.76	0.00	0.00	268.57
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	148.11	34.66	336.27	2,301.71
Anticipated crop nutrient removal	160.00	25.60	132.80	832.00
Actual crop nutrient removal	199.58	29.51	204.94	966.04
Nutrient balance	-51.47	5.15	131.32	1,335.67
Applied to removed ratio	0.74	1.17	1.64	2.38

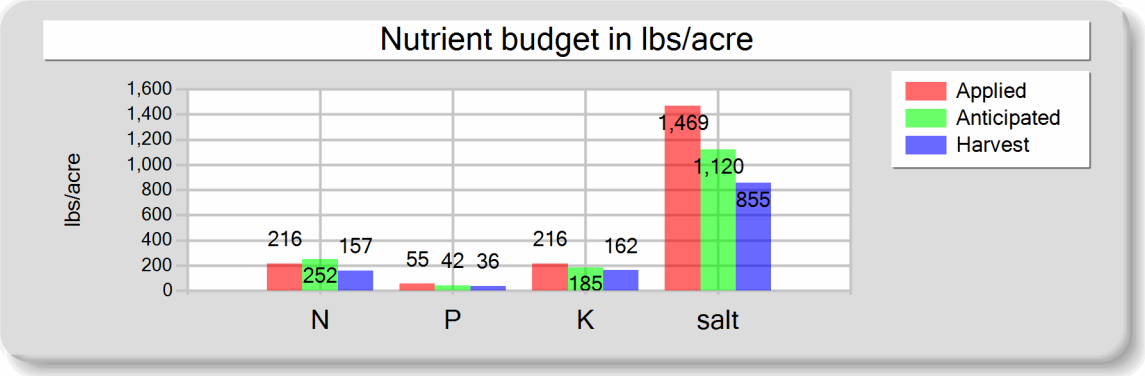
Fresh water applied	
20,580,000.00	*gallons*
757.89	*acre-inches*
8.06	*inches/acre*

Process wastewater applied	
5,760,000.00	*gallons*
212.12	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
2	*harvests*

Fd #6 - 06/15/2023: Corn, silage

Field name: Fd #6 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	159.42	48.32	174.19	0.00
Process wastewater	38.97	6.37	41.83	255.68
Fresh water	12.46	0.00	0.00	1,212.96
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	215.52	54.69	216.01	1,468.64
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	157.09	36.40	161.76	855.44
Nutrient balance	58.42	18.29	54.26	613.19
Applied to removed ratio	1.37	1.50	1.34	1.72

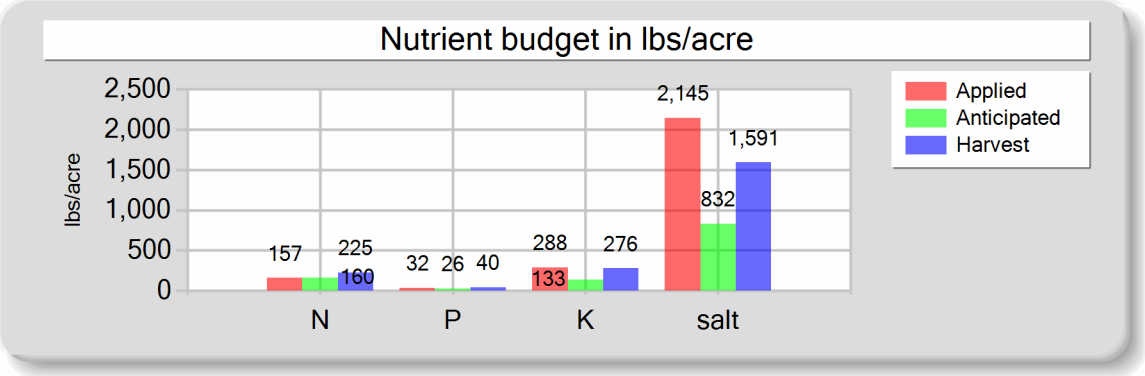
Fresh water applied	
92,946,000.00	*gallons*
3,422.89	*acre-inches*
36.41	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
0.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #7 - 10/28/2022: Oats, silage-soft dough

Field name: Fd #7 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	146.95	32.07	288.40	1,873.25
Fresh water	2.79	0.00	0.00	271.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.73	32.07	288.40	2,144.53
Anticipated crop nutrient removal	160.00	25.60	132.80	832.00
Actual crop nutrient removal	224.92	39.68	276.12	1,590.92
Nutrient balance	-68.19	-7.61	12.28	553.61
Applied to removed ratio	0.70	0.81	1.04	1.35

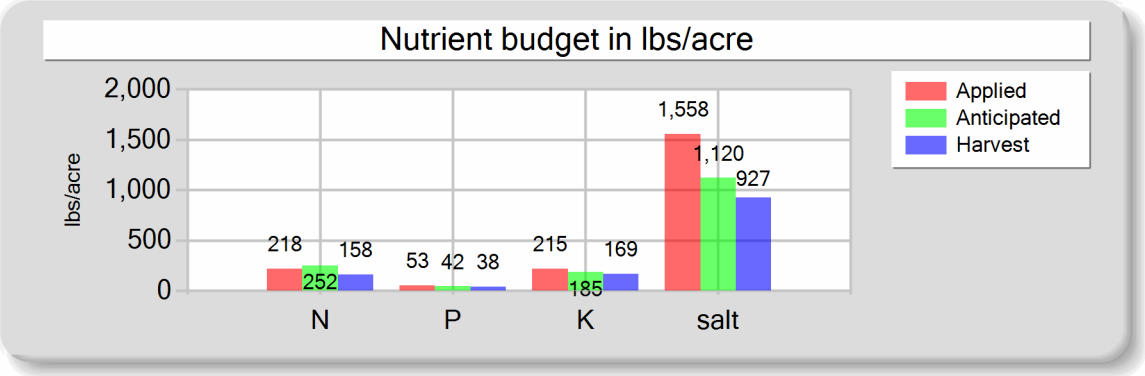
Fresh water applied	
15,480,000.00	*gallons*
570.08	*acre-inches*
8.14	*inches/acre*

Process wastewater applied	
3,456,000.00	*gallons*
127.27	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fd #7 - 06/11/2023: Corn, silage

Field name: Fd #7 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	145.76	44.18	159.26	0.00
Process wastewater	52.32	8.55	56.17	343.34
Fresh water	12.47	0.00	0.00	1,214.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.56	52.73	215.43	1,557.78
Anticipated crop nutrient removal	252.00	42.00	184.80	1,120.00
Actual crop nutrient removal	158.45	38.43	168.56	927.09
Nutrient balance	59.11	14.30	46.86	630.70
Applied to removed ratio	1.37	1.37	1.28	1.68

Fresh water applied	
69,300,000.00	*gallons*
2,552.08	*acre-inches*
36.46	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral solids

Sample and source description: Corral solids

Sample date: 06/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	4,910.00	17,700.00							0.00
DL	1,000.00	1,000.00	1,000.00							0.10

Corral solids

Sample and source description: Corral solids

Sample date: 11/07/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 35.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	9,210.00	63,200.00							0.00
DL	1,000.00	1,000.00	1,000.00							0.10

B. PROCESS WASTEWATER ANALYSES

LG-1-4,2022

Sample and source description: LG-1-4,2022

Sample date: 11/17/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	130.00	87.00	0.00	0.00	52.70	594.00								3,400.00	3,120
DL	0.80	0.50	0.50	0.50	0.50	2.00								1.00	10

LG-1-1

Sample and source description: LG-1-1

Sample date: 03/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	470.00	277.00	0.00	0.00	90.40	753.00								4,970.00	5,260
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

LG-1-2

Sample and source description: LG-1-2

Sample date: 06/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	224.00	126.00	0.00	0.00	54.40	371.00								3,140.00	1,440
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

LG-1-3

Sample and source description: LG-1-3

Sample date: 08/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	157.00	89.60	0.00	0.00	7.89	38.00								2,080.00	1,060
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

LG-1-4

Sample and source description: LG-1-4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	339.00	241.00	0.00	0.00	102.00	519.00								4,650.00	3,200
DL	0.50	0.50	0.50	0.50	0.50	2.00								1.00	10

C. FRESH WATER ANALYSES

Barn Dom.

Domestic Barn

Sample description: Domestic Barn

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.00	0.00	6.00								333.00	213
DL	0.50	0.50	0.50								1.00	10

D-1

Domestic Well #1

Sample description: Domestic Well #1

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.45	0.00	3.45								297.00	190
DL	0.50	0.50	0.50								1.00	10

D-4

Domestic Well #4

Sample description: Domestic Well #4

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.38	0.00	4.38								305.00	195
DL	0.50	0.50	0.50								1.00	10

D-5

Domestic Well #5

Sample description: Domestic Well #5

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.89	0.00	3.89								302.00	193
DL	0.50	0.50	0.50								1.00	10

D-6

Domestic Well #6

Sample description: Domestic Well #6

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.80	0.00	0.80								118.00	76
DL	0.50	0.50	0.50								1.00	10

D-7

Domestic Well #7

Sample description: Domestic Well #7

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.66	0.00	0.66								119.00	76
DL	0.50	0.50	0.50								1.00	10

DW-1

DW-1

Deep Well #1

Sample description: Deep Well #1

Sample date: 08/23/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.75	0.00	8.75								345.00	221
DL	0.50	0.50	0.50								1.00	10

River

Merced River

Sample description: Merced River

Sample date: 11/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.51	0.00	1.51								230.00	147
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Fd #1,2 - 10/25/2022: Oats, silage-soft dough

Fd #1,2 - 10/25/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,250.00	15,400.00		9.60
DL	1,000.00	1,000.00	1,000.00		0.10

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/25/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,500.00	2,700.00	11,500.00		10.70
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #1,2 - 06/30/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 10/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	1,820.00	9,630.00		3.10
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #1,2 - 06/30/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 10/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,100.00	1,830.00	10,300.00		3.60
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #11 - 01/02/2016: Pasture Grass

Pasture Grass

Sample and source description: Pasture Grass

Sample date: 08/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	2,680.00	15,400.00		19.50
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #1a/2a - 01/02/2016: Pasture Grass

Pasture Grass

Sample and source description: Pasture Grass

Sample date: 08/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,500.00	2,680.00	15,400.00		19.50
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #3 - 10/26/2022: Oats, silage-soft dough

Winter Forage #6A

Sample and source description: Winter Forage #6A

Sample date: 05/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	2,930.00	20,100.00		11.80
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #3 - 06/15/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 10/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,010.00	10,400.00		3.20
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #6 - 10/27/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	3,020.00	21,700.00		9.70
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #6 - 10/27/2022: Oats, silage-soft dough

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,500.00	2,700.00	11,500.00		10.70
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #6 - 06/15/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,340.00	10,400.00		5.50
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #7 - 10/28/2022: Oats, silage-soft dough

Winter Forage #7A

Sample and source description: Winter Forage #7A

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,170.00	15,100.00		8.70
DL	1,000.00	1,000.00	1,000.00		0.10

Fd #7 - 06/11/2023: Corn, silage

Corn Silages

Sample and source description: Corn Silages

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,280.00	10,000.00		5.50
DL	1,000.00	1,000.00	1,000.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

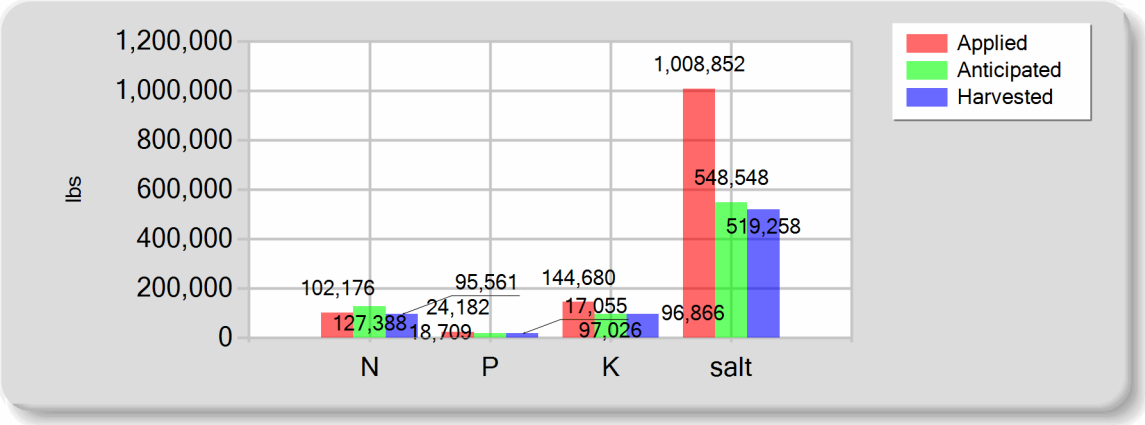
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

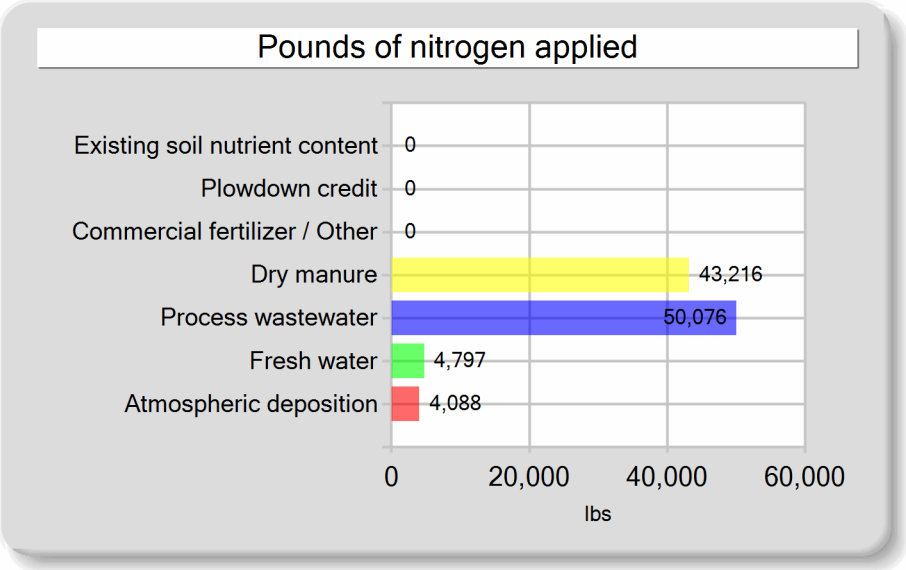
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

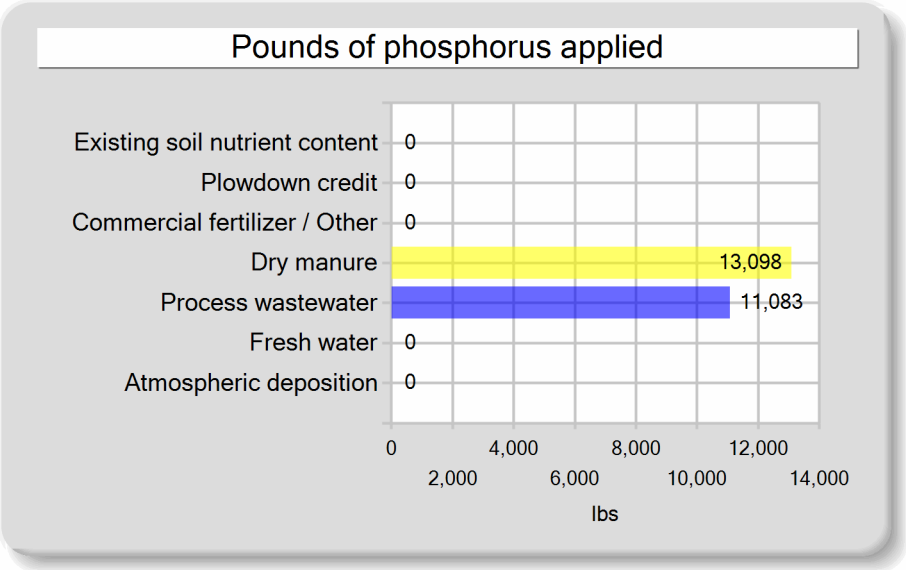
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	43,216.22	13,098.25	47,217.72	0.00
Process wastewater	50,075.61	11,083.36	97,461.86	592,171.88
Fresh water	4,796.59	0.00	0.00	416,680.10
Atmospheric deposition	4,088.00	0.00	0.00	0.00
Total nutrients applied	102,176.42	24,181.61	144,679.58	1,008,851.98
Anticipated crop nutrient removal	127,388.00	18,708.80	97,026.40	548,548.00
Actual crop nutrient removal	95,561.31	17,055.11	96,865.85	519,258.18
Nutrient balance	6,615.11	7,126.50	47,813.73	489,593.80
Applied to removed ratio	1.07	1.42	1.49	1.94

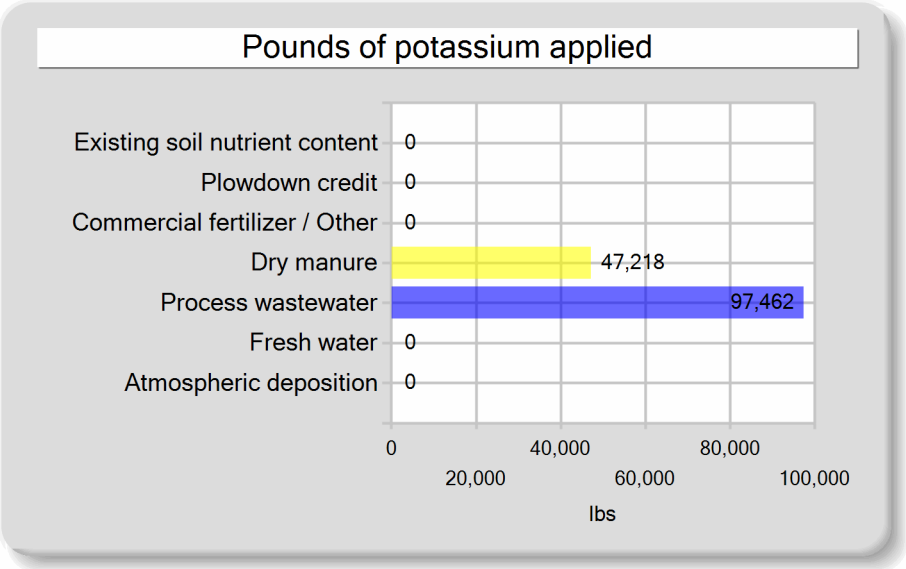
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

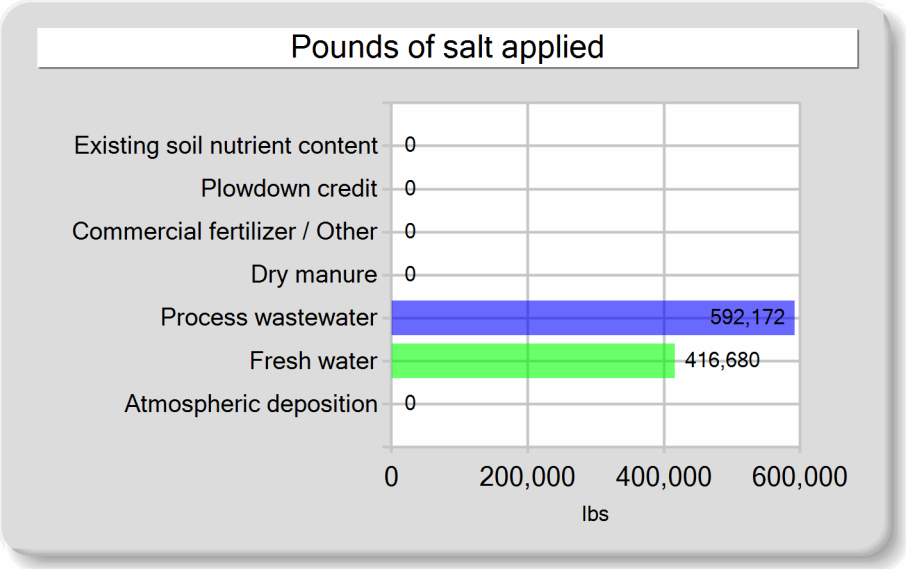


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

PH Ranch 03:

Manures from dairy are composted by the owners, Corral solids were applied to Corn silage crops and the rest is being held in storage.

Fields 5, 8, 9, 10 and 14 are all leased to a Sand and Gravel company and are not farmed at this time.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Michael D. Bettencourt (Jun 27, 2024 14:35 PDT)	Paul van Warmerdam (Jun 27, 2024 14:12 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bettencourt Family Property	Paul Van Warmerdam
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	Jun 27, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
PH Ranch 03
11570 N. Shaffer Rd.
Winton, CA 95388

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 2:30 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 23.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-1	T3235034B	Electrical Conductivity (EC)	EPA 120.1 and	0.345	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.75	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	221	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	101	95.1 (T3235024B)	95.9 (T3235024B)	0.815
Electrical Conductivity (EC)	97.1			4.81
Nitrate Nitrogen (NO3-N)	104			7.24

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: hlcdan@yahoo.com

Client: PH RANCH 03
Mailing address: 11570 N. Shaffer Rd. Winton CA 95388
(not facility address)
Phone#: 209-617-3388

Sampled By: Daniel Roos
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	8-23-23	0712	LG-1-3		23.8C			X			
2	8-23-23	0720	DW-1		23.6C	RUNNING	X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st		HeartLand Consulting		8-23-23 1430		
2nd		DENELE ANALYTICAL, INC.	8-23-23 2:30			T3235034

Required to be submitted to any agency? (YES / NO)

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Compliance Analysis Report

Customer
PH Ranch 03
11570 N. Shaffer Rd.
Winton, CA 95388

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Barn-Dom	T3334014A	Electrical Conductivity (EC)	EPA 120.1 and	0.333	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.00	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	213	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
PH Ranch 03
11570 N. Shaffer Rd.
Winton, CA 95388

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D-4	T3334014B	Electrical Conductivity (EC)	EPA 120.1 and	0.305	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.38	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	195	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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Compliance Analysis Report

Customer
PH Ranch 03
11570 N. Shaffer Rd.
Winton, CA 95388

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D-5	T3334014C	Electrical Conductivity (EC)	EPA 120.1 and	0.302	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.89	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	193	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
PH Ranch 03
11570 N. Shaffer Rd.
Winton, CA 95388

Grower

Consultant:
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D-1	T3334014D	Electrical Conductivity (EC)	EPA 120.1 and	0.297	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.45	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	190	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D-6	T3334014E	Electrical Conductivity (EC)	EPA 120.1 and	0.118	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.797	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	75.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D-7	T3334014F	Electrical Conductivity (EC)	EPA 120.1 and	0.119	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.664	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	76.0	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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Consultant:
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P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 11/30/23 2:05 PM
SUBMITTED BY: Dan
ANALYZED DATE: 12/5/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
River	T3334014G	Electrical Conductivity (EC)	EPA 120.1 and	0.230	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.51	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	147	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	97.0			0.106
Nitrate Nitrogen (NO3-N)	95.0	101 (T3334011D)	100 (T3334011D)	0.756

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: PH RANCH 03
Mailing address: 11570 N. Shaffer Rd. Winton CA 95388
(not facility address)
Phone#: 209-617-3388

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	11-30-23	655	BARN-DOM		14.2°C	710 MIN	X				
2	↓	705	D-4		14.0°	"	X				
3		710	D-5		14.6°	"	X				
4		715	D-1		14.0°	"	X				
5		725	D-6		14.0°	"	X				
6		730	D-7		14.8°	"	X				
7	11-30-23	740	RIVER		19.0°	–	X				
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	D Roos	HeartLand Consulting		11-30-23 1405		
2nd		DENELE ANALYTICAL, INC.	11-30-23 2:05			T334014

Required to be submitted to any agency? (YES / NO)

2023 AR PH RANCH 03

Final Audit Report 2024-06-27

Created:	2024-06-27
By:	Daniel Roos (hlcdan@yahoo.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA5rM4hT3ABfCMxi-5-1xHx2nKsSWnuc5a

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